

# Chemistry — Assignment 11: Stability of Carbanion & Free Radical

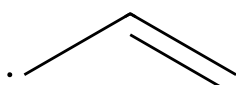
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Chemistry (Foundational Chemistry) · Semester 5

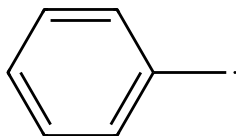
**Question.** Explain the stability of a carbanion and of a free radical (inductive, resonance, alkyl/benzyl).

**Carbanion:**  $-I$  (electron-withdrawing) groups *increase* stability;  $+I$  alkyl groups *destabilise*, so the alkyl order is methyl  $> 1^\circ > 2^\circ > 3^\circ$  (opposite to carbocation). Resonance strongly stabilises — allyl and benzyl carbanions delocalise into the  $\pi$  system.

**Free radical:**  $+I$  alkyl groups and hyperconjugation *increase* stability  $\rightarrow 3^\circ > 2^\circ > 1^\circ > \text{methyl}$ ; resonance makes allyl and benzyl radicals especially stable.



allyl



benzyl